Nested Association Mapping (NAM) Reveals Genetic Architecture of Agronomic Traits in Quinoa

Reviewer 1

The manuscript presents a comprehensive and well-designed study aimed at dissecting the genetic architecture of key agronomic traits in quinoa through the integration of nested association mapping (NAM), genome-wide association studies (GWAS), and genomic prediction. The topic is timely and highly relevant, considering the increasing global interest in quinoa as a climate-resilient crop with exceptional nutritional value.

Response: Dear Reviewer, thank you for your comments and for your positive evaluation of our work. We appreciate the time and effort you devoted to reviewing our manuscript. Please find our detailed response to your question below.

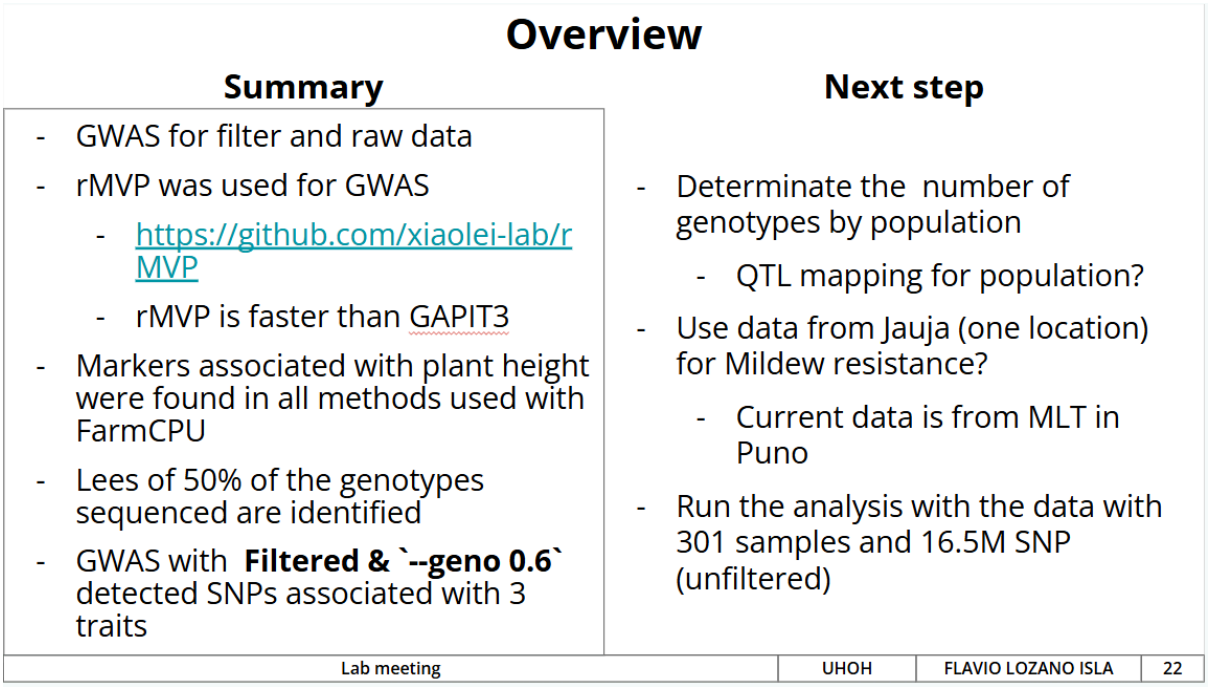
Comment 1: I have only one minor question for the authors. Did the authors compare the performance of different GWAS models implemented in GAPIT (e.g., GLM, MLM, CMLM, MLMM, BLINK, or others) before selecting FarmCPU? If so, it would be valuable to briefly explain the criteria used for choosing FarmCPU as the primary model and whether the significant loci identified by FarmCPU were consistent with those detected by other models.

Response 1: Thank you for this suggestion. We indeed performed several comparisons before selecting the final GWAS framework.

First, we compared the performance of two GWAS software packages, **GAPIT** (Lipka et al., 2012; Wang & Zhang, 2021) and **rMVP** (Yin et al., 2021). Both implementations produced virtually identical association results, identifying the same significant loci across the evaluated traits. Because rMVP provides faster computation and supports parallel processing, it was selected for the final analysis. A comparison of the outputs obtained from GAPIT and rMVP was previously presented during our laboratory meeting (attached figure).

We also evaluated the **NAM** R package, which is specifically designed for Nested Association Mapping populations. However, under our dataset and analytical conditions, this package identified fewer significant associations and showed lower consistency compared with the rMVP implementation. These comparisons are presented in the manuscript at **Supplementary Figures S6 and S7**.

Within rMVP, we further compared the performance of different GWAS models. Among them, **FarmCPU** consistently identified the same significant SNPs across multiple related traits while providing an effective balance between statistical power and control of false-positive associations. This performance is consistent with previous studies demonstrating that FarmCPU improves statistical power by iteratively fitting fixed and random effect models, thereby reducing confounding between testing markers and kinship while minimizing both false positives and false negatives (Liu et al., 2016). For these reasons, FarmCPU was selected as the primary GWAS model for the analyses presented in the manuscript.



References:

Lipka, A. E., Tian, F., Wang, Q., Peiffer, J., Li, M., Bradbury, P. J., ... & Zhang, Z. (2012). GAPIT: Genome Association and Prediction Integrated Tool. Bioinformatics, 28(18), 2397–2399. https://doi.org/10.1093/bioinformatics/bts444

Liu, X., Huang, M., Fan, B., Buckler, E. S., & Zhang, Z. (2016). Iterative usage of fixed and random effect models for powerful and efficient genome-wide association studies. PLoS Genetics, 12(2), e1005767. https://doi.org/10.1371/journal.pgen.1005767

Wang, J., & Zhang, Z. (2021). GAPIT Version 3: Boosting power and accuracy for genomic association and prediction. Genomics, Proteomics & Bioinformatics, 19(4), 629–640. https://doi.org/10.1016/j.gpb.2021.08.005

Yin, L., Zhang, H., Tang, Z., Xu, J., Yin, D., Zhang, Z., ... & Liu, X. (2021). rMVP: A Memory-efficient, Visualization-enhanced, and Parallel-accelerated tool for Genome-Wide Association Study. Genomics, Proteomics & Bioinformatics, 19(4), 619–628. https://doi.org/10.1016/j.gpb.2020.10.007

Nested Association Mapping (NAM) Reveals Genetic Architecture of Agronomic Traits in Quinoa

Reviewer 2

This submission dissects the genetic architecture of major agronomic traits in quinoa by nested association mapping (NAM). The manuscript is well composed, but I wonder as to why the pages in the .docx file were not continuously or page-wise numbered. Besides, there are several spellings as well as syntax mistakes, e. g., 'planicle' in the last but the penultimate sentence in 'Abstract'. Anyway, those can be addressed to while copy-editing if the manuscript is finally accepted for publication in Sci Rep. However, my major concerns lie somewhere being listed as follows.

Response: We thank you for the evaluation of our manuscript and for the comments and suggestions. In the revised version, we have incorporated continuous page numbering and added line numbers throughout the manuscript to facilitate the review process. We have also revised the manuscript to correct the typographical, spelling, and syntax errors. All modifications have been highlighted in red in the revised manuscript for ease of reference.

Comment 1 Abstract: "To address this gap, two nested association mapping (NAM) populations were developed using the recurrent parents Salcedo INIA, a high-yielding cultivar, and Kancolla, a traditional highland landrace, crossed with four diverse founder cultivars from the Peruvian Altiplano. A total of 125 recombinant inbred lines were evaluated under multi-environment field trials in the Peruvian highlands for phenology, plant architecture, and grain yield traits." Now, please refer to M & M sub-section 2.2: "Phenotypic data were obtained from our previous study [24], based on multiyear field trials conducted at two experimental sites in the highlands of the province of Puno, Peru, near Lake Titicaca and approximately 40 km apart: Camacani (3,825 m a.s.l.) and Illpa (3,820 m a.s.l.)." I find it highly ambiguous because the preceding sentence clearly states that the entire phenotyping was not done in the submitted study (paper), rather phenotypes scored earlier in their 2023 Euphytica (219: 26) paper ("Enhancing quinoa cultivation in the Andean highlands of Peru: a breeding strategy for improved yield and early maturity adaptation to climate change using traditional cultivars"; Ref. No. 24 in the Ref. List) were used here to perform the NAM.

Response 1: We thank the reviewer for this important observation. We agree that the original wording could be interpreted as if the phenotypic evaluations had been conducted specifically for the present study. To avoid this ambiguity, we have revised the Abstract (Line 46-63), Introduction (Line 103-105), and Materials and Methods (Line 110-114) to explicitly state that the phenotypic data were previously generated and published by Lozano-Isla et al. (2023). We now clearly indicate that the present study integrates these published multi-environment phenotypic data with newly generated whole-genome resequencing data to perform population structure analyses, GWAS, candidate gene identification, and genomic prediction. We also emphasize that all genome-wide analyses presented in this study, including SNP discovery, population structure analysis, GWAS, candidate gene identification, and genomic prediction, were conducted specifically for the present study and were not reported in the previous publication.

Comment 2: Similarly, please refer to M & M section 2.1 describing plant material and population development, with a figure (Fig. 1). The same was more elaborately described in Table 1 of their above-cited Euphytica paper. Though the authors have adequately cited their earlier Euphytica work (Ref. No. 24 in the Ref. List), I find M & M sub-sections 2.1 and 2.2 REDUNDANT here including the phenotypic data analyses vide the last paragraph of the M & M sub-section 2.2. The referred

mixed-model analyses for the given traits, except for stem diameter (cm), were earlier reported in their Euphytica paper. In such circumstances, I cannot assuage whether the phenotypic analyses being reported in the study exclusively pertain to the submitted study or to the ones earlier reported in their Euphytica paper but again being reported here. To my understanding it amounts to duplication.

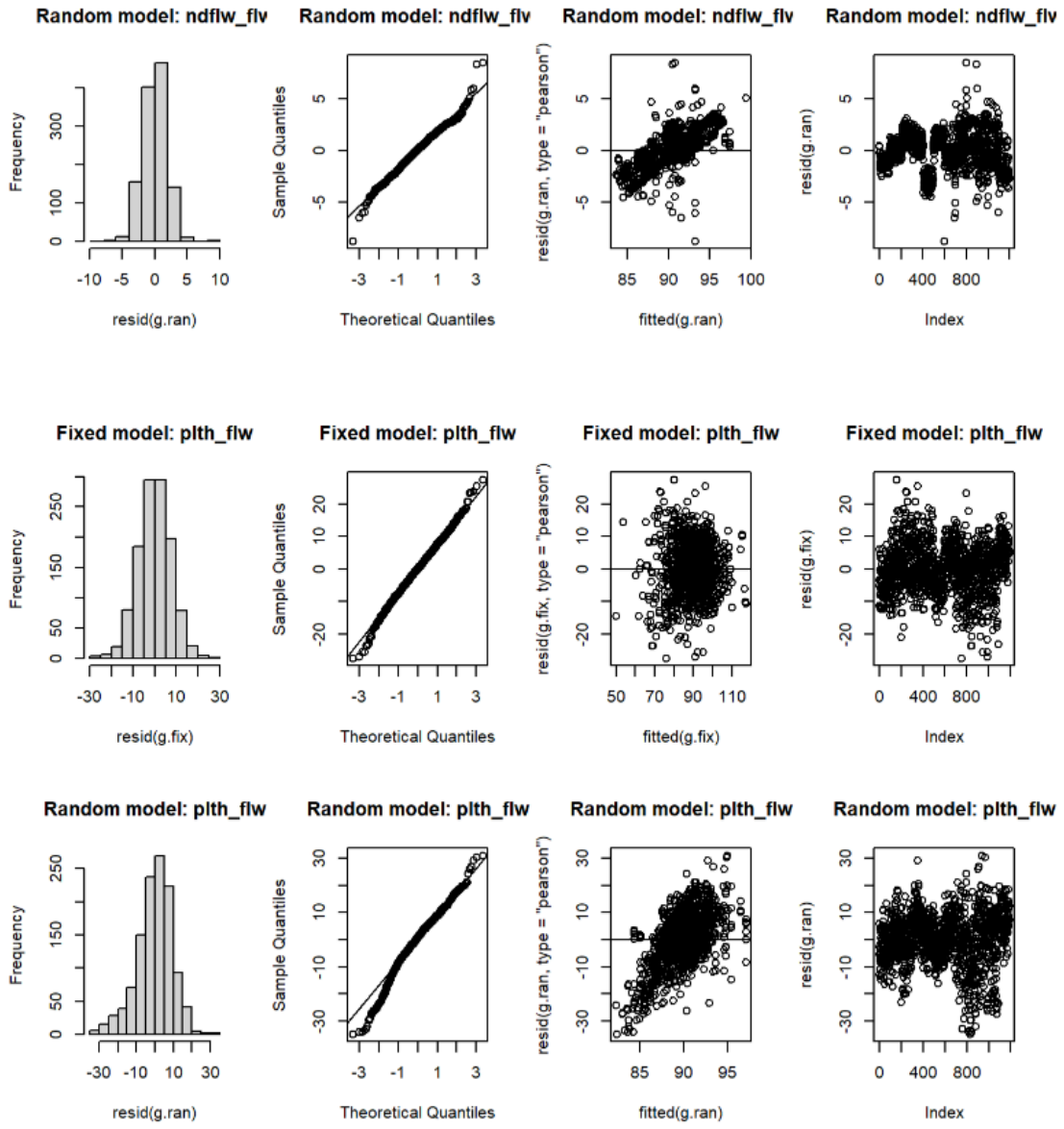
Comment 2: We thank the reviewer for raising this important concern. To avoid any ambiguity, we revised Sections 2.1 and 2.2. The descriptions of the population development, field trials, and mixed-model phenotypic analyses have been considerably shortened and now explicitly refer readers to Lozano-Isla et al. (2023), where these methods were originally reported. We further clarify that the published BLUE values from that study were used as phenotypic inputs for the genomic analyses presented here. We also explicitly state that the genomic dataset and all downstream genomic analyses presented in this manuscript were generated and performed specifically for the present study and were not reported in Lozano-Isla et al. (2023). Lines (144 - 146)

Question 3: By the way, were the predictor (response) variables, such as plant height (cm), stem diameter (cm), panicle length (cm), panicle width (cm), grain yield per plant (g), yield per plot, etc. were tested for homoscedasticity (normality and homogeneity of error variance) prior to performing the mixed-model analyses? Nowhere could I find it mentioned in the manuscript. Data analysing by parametric statistics must obey the homoscedasticity assumptions. Authors should check and report the results in the paper. If any response variable violated the assumptions, it must appropriately transformed prior to performing the downstream phenotypic analyses.

Response 3: We thank the reviewer for this comment. The mixed-model analyses used to obtain the phenotypic BLUEs were not performed as part of the present study but were previously conducted and published in Lozano-Isla et al. (2023) at Supplementary file 3 section 5.3. During the fitting of the mixed models, all phenotypic traits underwent a quality control procedure, including data cleaning, identification and removal of outliers, and verification of model assumptions through residual diagnostic analyses. As recommended by Kozak and Piepho (2017).

To improve the clarity of the manuscript, we have now explicitly stated in the Materials and Methods that these diagnostic procedures were performed prior to the mixed-model analyses in the previous study (Lines 157-161). In addition, we have included the distribution of all analyzed traits in Supplementary Figure S4, together with a statement indicating that the assumptions were evaluated before the statistical analyses. Since the present study uses the published BLUE estimates as phenotypic inputs for GWAS and genomic prediction, no additional mixed-model analyses were conducted here.

Example of the analysis for quality control procedure, including data cleaning, identification and removal of outliers, and verification of model assumptions through residual diagnostic analyses



Reference:

Kozak, M., & Piepho, H. (2017). What's normal anyway? Residual plots are more telling than significance tests when checking ANOVA assumptions. *Journal of Agronomy and Crop Science*, *204*(1), 86–98. https://doi.org/10.1111/jac.12220

Bernal-Vasquez, A., Utz, H., & Piepho, H. (2016). Outlier detection methods for generalized lattices: a case study on the transition from ANOVA to REML. Theoretical and Applied Genetics, 129(4), 787–804. https://doi.org/10.1007/s00122-016-2666-6

Comment 4: In view of point nos 1 and 2 as above, this study as being reported here left us only with genotyping followed by NAM and genomic prediction. Though authors have mentioned it sub-section 4.4 of the 'Discussion' section, resequencing at as mere ~1x coverage happens to be one of the most serious limitations of acquiring the genotypes, compounded further by a much lower population size. Nevertheless, "High-quality reads were aligned to the quinoa reference genome (QQ74; CoGe ID 60716) using BWA-MEM v0.7.17 [35]. " somewhat mitigates the issue.

Response 4: We thank the reviewer for this important observation. We agree that the sequencing depth (~1× coverage) represents one of the main limitations of the present study, as already acknowledged in the Discussion. The decision to use low coverage whole-genome resequencing was primarily driven by budgetary constraints, allowing us to genotype a larger number of recombinant inbred lines while maintaining adequate genome-wide marker coverage.

Although genotype imputation is frequently used to improve genotype calls from low coverage sequencing data, its accuracy strongly depends on the availability of a large, high quality, population specific reference panel with closely related haplotypes. Such reference resources are currently unavailable for quinoa, particularly for the Peruvian highland germplasm analyzed in this study. Under these circumstances, imputation could introduce additional genotype uncertainty if the available reference panel does not adequately represent the haplotypic diversity of the Peruvian highland germplasm. Therefore, we opted for a conservative analytical strategy based on directly observed SNPs after stringent quality filtering rather than relying on imputed genotypes.

To clarify this point, we have expanded the Discussion to explain the rationale for using low coverage sequencing and for not performing genotype imputation in the present study (Line 603-610).

Reference:

Watowich, M. M., Chiou, K. L., Graves, B., Montague, M. J., Brent, L. J. N., Higham, J. P., Horvath, J. E., Lu, A., Martinez, M. I., Platt, M. L., Schneider‐Crease, I. A., Lea, A. J., & Snyder‐Mackler, N. (2025). Best practices for genotype imputation from low‐coverage sequencing data in natural populations. Molecular Ecology Resources, 25(5), e13854. https://doi.org/10.1111/1755-0998.13854

Comment 5: "Variants with more than 50% missing genotypes were filtered out using the expression F_MISSING > 0.5."(sub-section 2.3). Is this not too high (50 % call rate)? Or, is it F_MISSING > 0.05 (95% call rate)? I could not follow it. With only 50% call rate, the statistical power would have been greatly reduced resulting in many false-positive association. Most importantly, rare variants (alleles) are greatly affected. I don't find that the missing genotypes were imputed. Authors are required to address this issue and preferably consider recalculation. Setting F_MISSING > 0.5 and performing the downstream NAM is the foremost bottleneck associated with the genotypes in this study. This renders me unable to assess the reliability of the associations being reported in this study. Further, I wonder as to why the authors did not bring into the attention in sub-section 4.4 (Limitations and perspectives) of the 'Discussion' section.

Response 5: We thank the reviewer for this important concern regarding genotype missingness and its potential impact on the downstream analyses. Following the reviewer's recommendation, we reprocessed the complete genomic dataset using a more stringent genotype call-rate threshold of 80% and repeated the downstream analyses using this revised dataset. The complete filtering procedure and resulting dataset are documented in ESM 1.

The application of the 80% call rate threshold produced only limited changes in genetic diversity estimates and population structure, and the genomic prediction results remained consistent with those obtained under the previous filtering scheme. However, because GWAS is particularly sensitive to marker availability and allele-frequency distributions, the GWAS analyses were recalculated using the revised SNP dataset. Consequently, the significant associations, candidate gene analyses, and related results were updated according to the new marker set.

We have revised the entire manuscript accordingly, including the Materials and Methods, Results, Discussion, Supplementary Tables and Figures, and Abstract and Conclusion, to ensure that all reported values and interpretations correspond to the 80% genotype call rate dataset. We have also clarified the filtering procedure in the Methods and explicitly acknowledged the implications of genotype completeness in the Discussion. (Line 177-183 and 608-610)

50% missing genotypes (F_MISSING > 0.5)	80% missing genotypes (F_MISSING > 0.8)

`> vcf <- read.vcfR("xclude/quinoa_0.5.vcf.gz")`
`Scanning file to determine attributes.`
`File attributes:`
`  meta lines: 51`
`  header_line: 52`
`  variant count: 61528`
`  column count: 357`
`Meta line 51 read in.`
`All meta lines processed.`
`gt matrix initialized.`
`Character matrix gt created.`
`  Character matrix gt rows: 61528`
`  Character matrix gt cols: 357`
`  skip: 0`
`  nrows: 61528`
`  row_num: 0`
`Processed variant: 61528`
`All variants processed`
	`> vcfile <- "xclude/quinoa_pops.vcf.gz"`
	`> snpdt <- snpR::read_vcf(file = vcfile)`
	`Scanning file to determine attributes.`
	`File attributes:`
	`  meta lines: 53`
	`  header_line: 54`
	`  variant count: 3863468`
	`  column count: 216`
	`Meta line 53 read in.`
	`All meta lines processed.`
	`gt matrix initialized.`
	`Character matrix gt created.`
	`  Character matrix gt rows: 3863468`
	`  Character matrix gt cols: 216`
	`  skip: 0`
	`  nrows: 3863468`
	`  row_num: 0`
	`Processed variant: 3863468`
	`All variants processed`

The table compares the VCF datasets obtained under two genotype missingness thresholds. Under the 50% missingness threshold 61,528 variants were retained, whereas under the 80% missingness threshold 3,863,468 variants were retained.

Comment 6: Why no LD decay map presented in the study? It's one of the most primary requirements of any population-based and/or AM work. I fail to understand.

Response 6: We thank the reviewer for this valuable suggestion. Following the reviewer's recommendation, we evaluated genome-wide linkage disequilibrium (LD) separately for the two NAM subpopulations (SAL and KAN) using PLINK 2.0. The analysis included a large number of SNP pairs at short physical distances, allowing a reliable estimation of LD. However, although a slight reduction in LD was observed within the first kilobase, r2 rapidly reached a low background level and remained constant across larger physical distances. Consequently, no biologically meaningful genome-wide LD decay pattern was observed.

This result is consistent with the characteristics of our study population. The quinoa material consists of advanced recombinant inbred lines (RILs) derived from a nested association mapping (NAM) population. The repeated generations of selfing used during RIL development accumulate historical

recombination events, reducing long-range linkage disequilibrium while increasing mapping resolution (Yu et al., 2008; Buckler et al., 2009). Because the LD analysis did not provide additional biological information beyond confirming the overall low LD of these populations, we decided not to include the LD decay figure in the revised manuscript. We have instead clarified this point in the Discussion (Line 630-636).

References:

Buckler, E. S., Holland, J. B., Bradbury, P. J., Acharya, C. B., Brown, P. J., Browne, C., Ersoz, E., Flint-Garcia, S., Garcia, A., Glaubitz, J. C., Goodman, M. M., Harjes, C., Guill, K., Kroon, D. E., Larsson, S., Lepak, N. K., Li, H., Mitchell, S. E., Pressoir, G., . . . McMullen, M. D. (2009). The genetic architecture of maize flowering time. Science, 325(5941), 714–718. https://doi.org/10.1126/science.1174276

Yu, J., Holland, J. B., McMullen, M. D., & Buckler, E. S. (2008). Genetic design and statistical power of nested association mapping in Maize. Genetics, 178(1), 539–551. https://doi.org/10.1534/genetics.107.074245

Comment 7: Table 1: the column heads ‘ho’ ‘he’ ‘fis’ are not correct acronyms, though the former two were rather correctly presented in the legend.

Response 7: Thank you for this observation. The abbreviations in Table 1 have been corrected and are now consistent with the definitions provided in the table legend.

General comment: In view of the above, I’m sorry I cannot recommend this paper in its present form for further consideration in Sci Rep. However, if the authors can address the above issues, avoids duplications (point 1 and 2 above) and recalculate the NAM using at least a >=70 genotype call (preferably >=80%, that is, 20% missing genotypes), they can resubmit this paper for further consideration.

Response: We thank the reviewer for the constructive and detailed evaluation of our manuscript. We have addressed the concerns raised regarding the provenance of the phenotypic data, redundancy with our previous publication, phenotypic quality control, low-coverage sequencing and genotype missingness. We have revised all sections of the manuscript accordingly and performed additional analyses where appropriate. The revised manuscript now includes these clarifications and analyses, and all changes are indicated in the new version.